

Lab 2: Agile Backlog Creation & Sprint Simulation in Jira

Problem Statement #02 — Automated Rubric Assignment Evaluator

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Jira project: Automated Rubric Evaluator (APP board) | 2 sprints completed

1. Epics and User Stories (Backlog)

The Lab 1 functional requirements (FR-001 to FR-005, NFR-001) were grouped into four Epics, each broken into user stories in the “As a / I want / So that” format, with High / Medium / Low priorities.

Spaces / Automated Rubric Evaluator (PES1UG24AM154)										
AREP board										
SummaryTimelineActive sprintsBacklogCalendarListFormsDevelopmentDocs										
Ask AI Search work Filter Group										
Work	Assignee	Reporter	Priority	Status	Resolution	Created	Updated	Due date		
AREP-1 Epic 1: Rubric Authoring & Weighting	Unassigned	mallela rishi	High	To Do	Unresolved	Oct 05, 2026, 10:48 AM	Oct 05, 2026, 10:48 AM	None		
AREP-5 Story 1.1: Author Ordered Rubric Criteria	Unassigned	mallela rishi	High	Done	Unresolved	Oct 05, 2026, 10:49 AM	Oct 05, 2026, 10:53 AM	None		
AREP-6 Story 1.2: Validate Weights Total 100%	Unassigned	mallela rishi	High	Done	Unresolved	Oct 05, 2026, 10:49 AM	Oct 05, 2026, 10:54 AM	None		
AREP-7 Story 1.3: Choose Partial-Credit Mode	Unassigned	mallela rishi	Medium	Done	Unresolved	Oct 05, 2026, 10:49 AM	Oct 05, 2026, 10:54 AM	None		
AREP-2 Epic 2: Test-Suite Binding & Validation	Unassigned	mallela rishi	High	To Do	Unresolved	Oct 05, 2026, 10:48 AM	Oct 05, 2026, 10:48 AM	None		
AREP-8 Story 2.1: Bind Tests to Automated Criteria	Unassigned	mallela rishi	High	Done	Unresolved	Oct 05, 2026, 10:49 AM	Oct 05, 2026, 10:54 AM	None		
AREP-9 Story 2.2: Block Publication on Invalid Bind...	Unassigned	mallela rishi	High	Done	Unresolved	Oct 05, 2026, 10:49 AM	Oct 05, 2026, 10:54 AM	None		
AREP-10 Story 2.3: Report All Validation Errors in On...	Unassigned	mallela rishi	Medium	Done	Unresolved	Oct 05, 2026, 10:49 AM	Oct 05, 2026, 10:54 AM	None		
AREP-3 Epic 3: Reference Dry Run & Publication	Unassigned	mallela rishi	Medium	To Do	Unresolved	Oct 05, 2026, 10:48 AM	Oct 05, 2026, 10:48 AM	None		
AREP-11 Story 3.1: Run Dry Run Against Reference S...	Unassigned	mallela rishi	Medium	Done	Unresolved	Oct 05, 2026, 10:49 AM	Oct 05, 2026, 10:54 AM	None		
AREP-12 Story 3.2: Review Dry-Run Marks and Conf...	Unassigned	mallela rishi	Medium	Done	Unresolved	Oct 05, 2026, 10:49 AM	Oct 05, 2026, 10:55 AM	None		
AREP-13 Story 3.3: Student Views Published Rubric	Unassigned	mallela rishi	Medium	Done	Unresolved	Oct 05, 2026, 10:49 AM	Oct 05, 2026, 10:55 AM	None		
AREP-4 Epic 4: Versioning, Reuse & Auditability	Unassigned	mallela rishi	Medium	To Do	Unresolved	Oct 05, 2026, 10:48 AM	Oct 05, 2026, 10:48 AM	None		

Figure 1. Backlog list view: four Epics with their user stories, priorities and status.

Epic progress panels

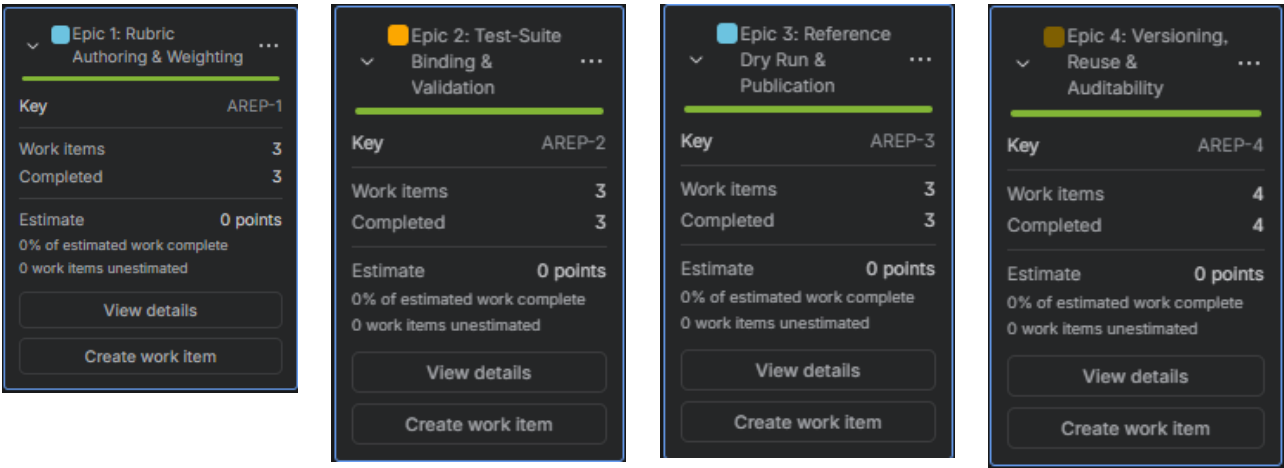


Figure 2. Epic panels showing work items and completion for Epics 1–4.

Epic and story mapping

Epic	Stories	Requirements
Epic 1: Rubric Authoring & Weighting	1.1 Author Ordered Rubric Criteria; 1.2 Validate Weights Total 100%; 1.3 Choose Partial-Credit Mode per Criterion	FR-001, FR-003
Epic 2: Test-Suite Binding & Dry Run	2.1 Bind Test Cases to Automated Criteria; 2.2 Block Publication on Unbound or Unknown Tests; 2.3 Dry Run Against Reference Solution	FR-002, FR-004
Epic 3: Rubric Publication & Student Access	3.1 Publish Rubric Version; 3.2 Student Views Rubric Version History; 3.3 Student Views Published Rubric	UC-01, UC-02, UC-07
Epic 4: Versioning, Reuse & Audit	4.1 Version and Lock Published Rubrics; 4.2 Create Revision of Locked Version; 4.3 Clone Rubric from Previous Offering; 4.4 Immutable Audit Record with Checksum	FR-005, NFR-001

2. Story Point Assignments

Stories were estimated with the Fibonacci scale (1, 2, 3, 5, 8) using Planning Poker reasoning: complexity, amount of work and uncertainty. Estimates by sprint:

Sprint	Story	Priority	Points
Sprint 1	1.1 Author Ordered Rubric Criteria	High	5
Sprint 1	1.2 Validate Weights Total 100%	High	3
Sprint 1	1.3 Choose Partial-Credit Mode per Criterion	Medium	3
Sprint 1	2.1 Bind Test Cases to Automated Criteria	High	5
Sprint 1	2.2 Block Publication on Unbound or Unknown Tests	High	3
Sprint 1	2.3 Dry Run Against Reference Solution	Medium	2
	Sprint 1 total		21
Sprint 2	3.1 Publish Rubric Version	High	8
Sprint 2	3.2 Student Views Rubric Version History	Medium	3
Sprint 2	3.3 Student Views Published Rubric	Medium	2
Sprint 2	4.1 Version and Lock Published Rubrics	Medium	5
Sprint 2	4.2 Create Revision of Locked Version	Medium	3
Sprint 2	4.3 Clone Rubric from Previous Offering	Low	3
Sprint 2	4.4 Immutable Audit Record with Checksum	High	5
	Sprint 2 total		29

Note: The Story Points field was not enabled on the Story screen of this Jira project, so Jira could not store these values and the burndown charts below use **Issue count** as the estimation statistic instead (Sprint 1 = 6 issues, Sprint 2 = 7 issues).

3. Sprint 1 (APP Sprint 1)

Goal: deliver rubric authoring, weight validation and test-suite binding with publication checks. Scope: APP-5 to APP-10 (6 stories), 1-week sprint.

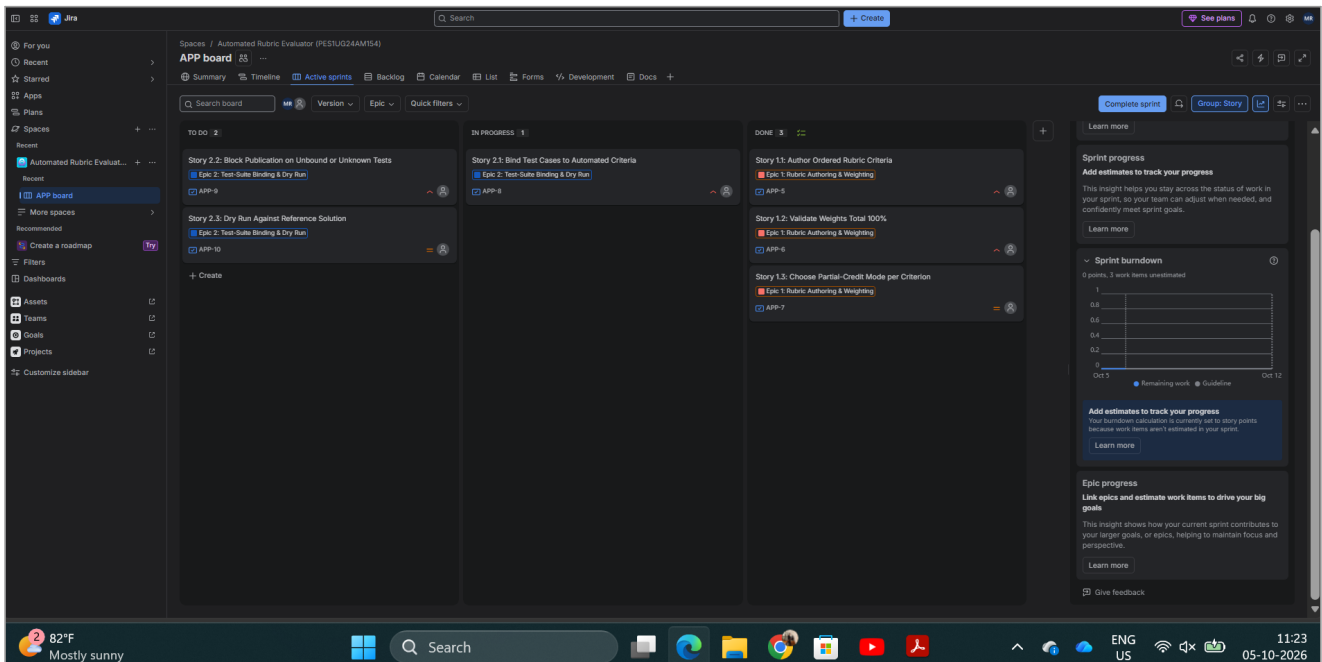


Figure 3. Active sprint board for Sprint 1 during execution (To Do, In Progress, Done).

Sprint 1 Burndown Chart

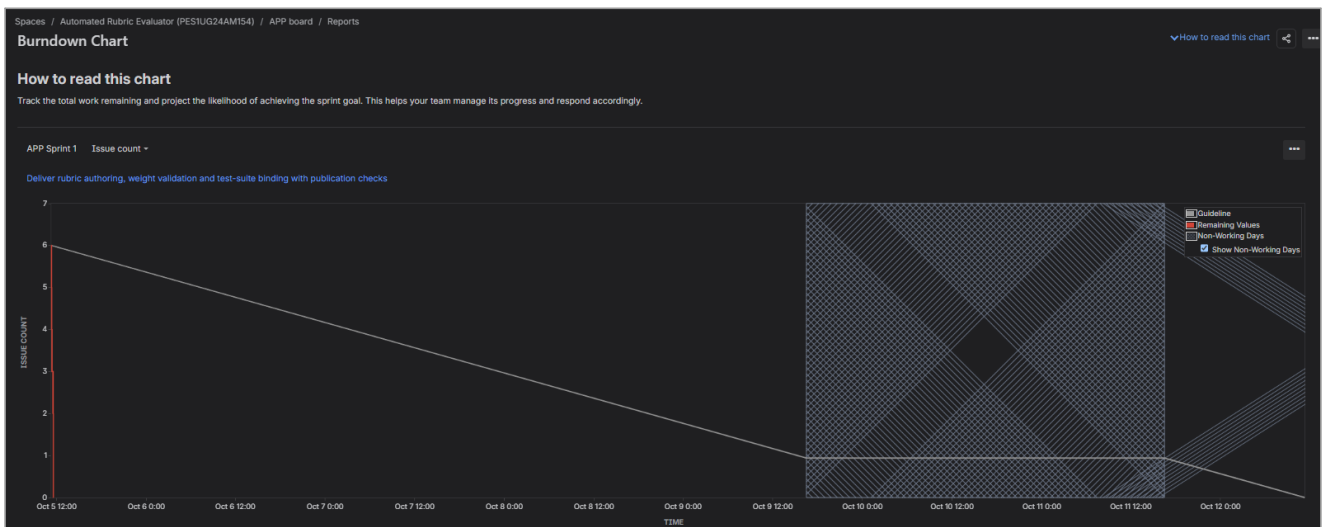


Figure 4. Burndown chart, APP Sprint 1 (estimation statistic: Issue count).

Date	Issue	Event Type	Event Detail	Inc.	Dec.	Remaining
05/Oct/26 11:18 AM	APP-10	Sprint start		1		
	APP-5			1		
	APP-6			1		
	APP-7			1		
	APP-8			1		
	APP-9			1		
	Total					6
05/Oct/26 11:20 AM	APP-5	Burndown	Issue completed	1	5	
05/Oct/26 11:21 AM	APP-6	Burndown	Issue completed	1	4	
05/Oct/26 11:22 AM	APP-7	Burndown	Issue completed	1	3	
05/Oct/26 11:30 AM	APP-8	Burndown	Issue completed	1	2	
05/Oct/26 11:33 AM	APP-8	Burndown	Issue reopened	1	3	
	APP-8	Burndown	Issue completed	1	2	
	APP-9	Burndown	Issue completed	1	1	
	APP-10	Burndown	Issue completed	1	0	
05/Oct/26 11:34 AM	APP-10	Sprint ended by maliela rishi				0
	APP-5					0
	APP-6					0
	APP-7					0
	APP-8					0
	APP-9					0
	Total					0

Figure 5. Sprint report events: sprint start with 6 issues, completions between 11:20 and 11:33, APP-8 reopened once, sprint ended 11:34 with 0 remaining.

4. Sprint 2 (APP Sprint 2)

Goal: deliver publication, student access, versioning, reuse and audit retention. Scope: APP-11 to APP-17 (7 stories), 1-week sprint.

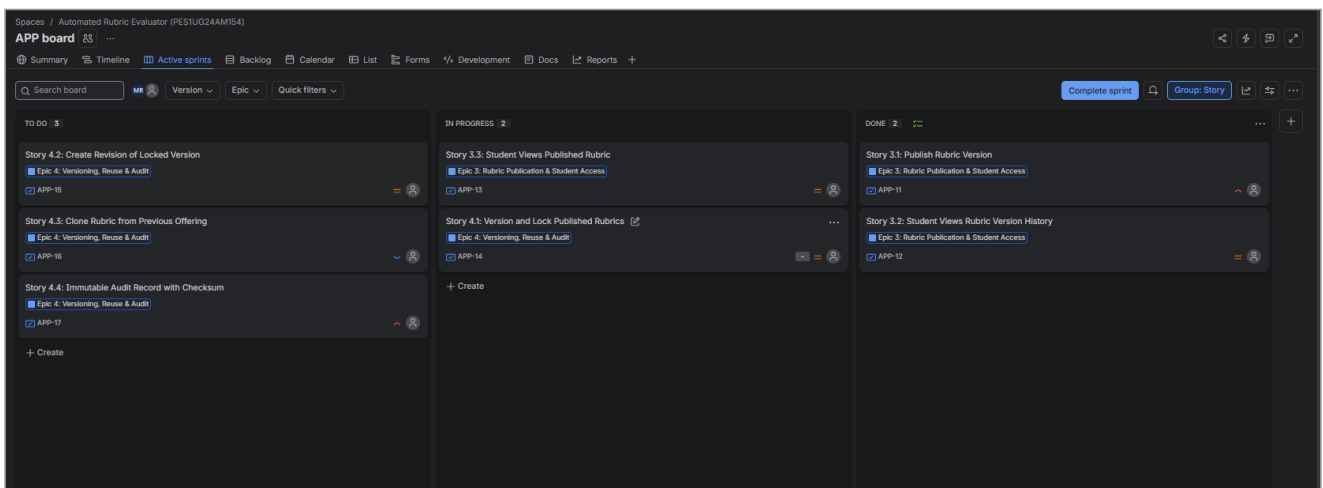


Figure 6. Active sprint board for Sprint 2 during execution.

Sprint 2 Burndown Chart

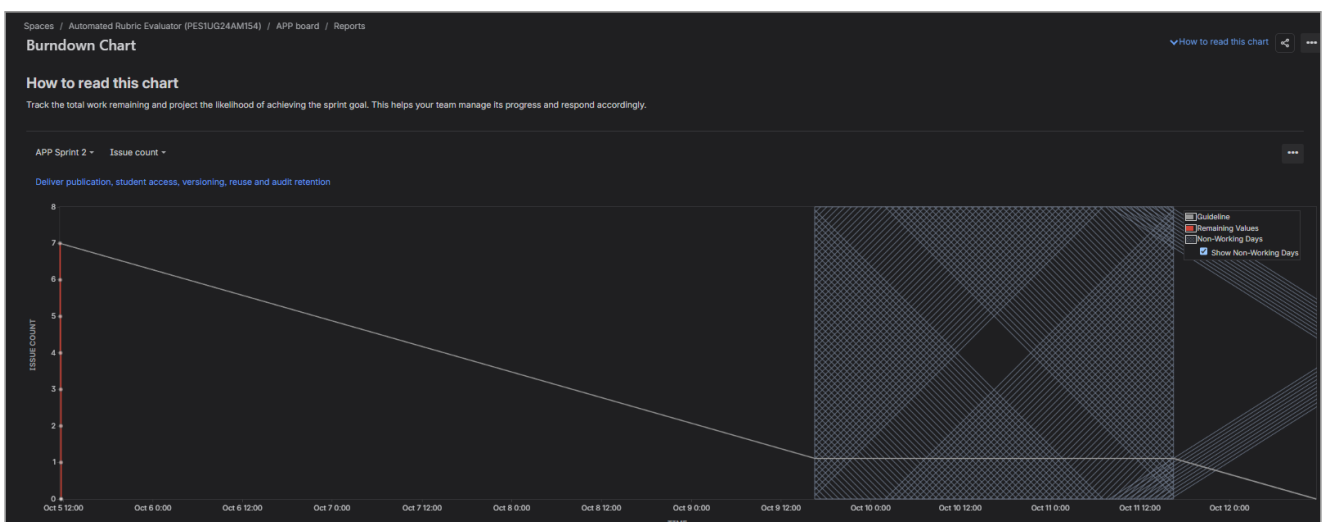


Figure 7. Burndown chart, APP Sprint 2 (estimation statistic: Issue count).

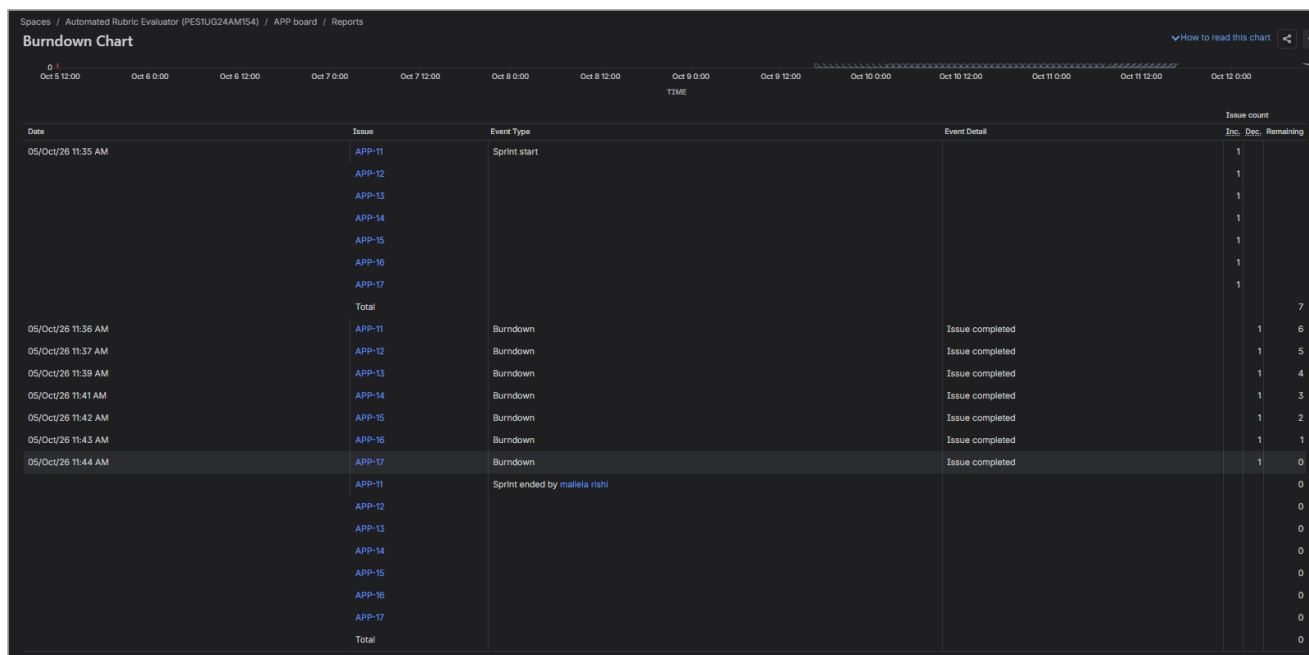


Figure 8. Sprint report events: 7 issues at start (11:35), one issue completed every 1–2 minutes from 11:36 to 11:44, 0 remaining at sprint end.

5. Reflection Questions

1. Did your estimations reflect the actual effort?

Only partly, and it cannot be fully verified. The stories were sized on the Fibonacci scale (21 points planned for Sprint 1, 29 for Sprint 2), with the heaviest estimate on Story 3.1, Publish Rubric Version (8), because it combines validation, dry-run confirmation and version creation. Because the sprint was simulated, each story took only a few minutes to move to Done, so the actual time spent says nothing about the estimates. Jira also had no Story Points field to record them, so progress was tracked by issue count. In a real sprint, the 8-point story would be the first candidate to split into smaller stories to reduce uncertainty.

2. Was your backlog well-prioritized?

Yes. Stories that protect correctness of grading (weights totalling 100%, binding tests to criteria, blocking publication on bad bindings, and the immutable audit record with checksum) were marked High, because each of those failures would be expensive or invisible until grades are published. Viewing stories and revision/versioning support were Medium, and cloning a rubric from a previous offering was Low because it saves effort but is not needed for the system to work. The sprints also follow the natural dependency order: authoring and binding (Sprint 1) before publishing, student access and version control (Sprint 2).

3. How did your simulated sprint align with your plan?

Both sprints finished all committed work with no scope added or removed: Sprint 1 started with 6 issues and ended with 0 remaining, and Sprint 2 started with 7 and ended with 0. The plan was a 1-week sprint, but the simulation compressed each sprint into 10–15 minutes. In Sprint 1, APP-8 was moved to Done, reopened at 11:33 and completed again within the same minute, which is the one deviation from a straight left-to-right flow. Sprint 2 was cleaner, with one story completed every 1–2 minutes from 11:36 to 11:44.

4. What insights did the burndown chart give about your team's capacity?

In both charts the red Remaining Values line stays below the grey Guideline for the whole sprint. The work finished almost immediately, so the line falls steeply near the start instead of tracking the guideline across the week. In a real sprint, finishing this early would suggest the team under-committed and could take on more. The numbers show a capacity of 6 issues (21 planned points) in Sprint 1 and 7 issues (29 planned points) in Sprint 2, which is a reasonable starting velocity of about 25 points for planning the next sprint. These figures come from a simulation, so they would need to be confirmed with real working time. The Sprint 2 chart also shows how one issue per completion produces a steady stepped burndown, which makes progress easy to read and is the shape a healthy sprint should have.